AI-Assisted Replication of Frye and Borisova (2019): Elections, Protest, and Trust in Government in a Competitive Autocracy

Brianne Anderson, Borna Atrchian, & Kassandra Gervasi.

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Introduction

Frye and Borisova (2019) investigate how elections and post-election protest shape political trust in a competitive autocracy, using a survey in Moscow conducted from November 25 to December 24, 2011 as a natural experiment. Their identification strategy exploits the exogenous assignment of interview dates across three periods — pre-election (before December 4), post-election (December 5-9), and post-protest (after December 10) — to causally attribute differences in reported trust indices to exposure to the December 4 parliamentary election and the December 10 Bolotnaya Square protest. The paper's central and somewhat counterintuitive finding is that exposure to the protest increased trust in government, especially among opposition voters who updated their beliefs upon observing the government's decision to permit rather than suppress the demonstration. The election, by contrast, had little effect, consistent with electoral fraud being "priced in" as non-novel information.

This project undertakes a structured replication and critical evaluation of these findings using two methodological approaches. The first, a full-AI round, tasked Claude Terminal with independently replicating the paper's results using only the original dataset without accessing the paper, then comparing those results to the published findings. The second, an AI-guided round, repeated the replication but layered in a number of additional robustness checks. Together, the two rounds allow us to assess not only if the paper's core results replicate, but the robustness of those results to increased methodological scrutiny and any limitations of statistical power, balance, causal identification, and analytical choices. The replication confirms the paper's main findings while uncovering meaningful concerns about subgroup power, anticipation contamination in Period 2, and the fragility of several institutional trust results once multiple comparisons corrections are applied.

Methods

The replication followed a two-track structure. To begin, the full-AI round provided Claude Terminal with only the original dataset and analysis code, without accessing the published paper, and prompted it to independently reproduce the paper's findings. It was subsequently given access to the published paper and prompted to produce a comparison report between its results and the paper's. The AI-assisted round repeated the replication with the same inputs, but added additional checks prompted by the research team, including code integrity verification, unit testing, balance diagnostics, causal identification tests, multiple comparisons corrections, and several new sensitivity analyses. Upon seeing errors in the initial AI-assisted results, we proceeded with a third and final Corrected AI-guided round. The replication began with a newly-created Claude account, and the memory was cleared between rounds to limit contamination of the results. Both rounds used R as the primary statistical environment.

Data and Variable Construction

The raw data file is a Stata, containing the Moscow Population Survey (N = 1,550), fielded November 25 to December 24, 2011 by the Levada Center. Several variables were originally in Russian and required translation. Period indicators were constructed to replicate the original Stata files, resulting in a pre-election period (Period 1: November 25-December 3), a post-election period (Period 2: December 5-9), and a final post-protest period (Period 3: December 11-25). Election Day (December 4) and Protest Day (December 10) were coded as separate binary indicators and excluded from the period dummies acting as the main regressors.

Trust in government was measured across eight state institutions on a 1-5 scale. A composite Government Trust Index was then constructed using the unweighted average of the institutional trust items, retaining respondents who answered at least one of eight sub-items. This is a departure from the original Stata code's listwise deletion on the composite, which produced a sample size of N= 1,232 for that outcome versus N = 1,521 in the replication. This discrepancy was documented and shown to have no effect on any substantive conclusion. Trust in the United Nations was retained as a separate placebo outcome and excluded from the government trust index. Partisan affiliation was then recoded into three groups consisting of United Russia (UR) supporters, apolitical respondents (non-voters and undecided), and the omitted opposition reference category, as well as interaction terms with each period dummy in the Table 3 models.

Estimation

All main models replicate the OLS specifications in Frye and Borisova (2019), with heteroskedasticity-consistent (HC1) robust standard errors and fixed effects for Moscow's ten administrative districts (okrugs), plus controls for wealth and education of respondents. The regressions of interest are period3 (Post-Protest), period2min (Post-Election), protestday, and electday, with Period 1 acting as the reference category. The estimand is the effect of protest exposure on expressed institutional trust, conditional on the covariates and district fixed effects. The partisan impact models in Table 3 add interaction terms for UR supporters and apolitical respondents crossed with each post-period dummy, allowing event effects to vary by party affiliation.

Unit Testing and Code Integrity

A unit-testing step was added in the third round, which was AI-guided, after a labeling error was identified in rounds 1 and 2 of the replication. Period labels in these rounds had been assigned to the wrong groups in many of the t-test comparisons and balance checks. Specifically, period1min == 0 had been labeled as “Period 1” when it actually captured Period 2 respondents, reversing the direction of effects in descriptive comparisons. The Corrected AI-guided round unit tests verified the validity of each period indicator by confirming that the dates associated with each dummy matched the expected calendar windows, and confirmed that all analyses produced means and differences consistent with the published paper. Additional code integrity checks verified that all trust variables fell within the valid 1-5 scale range after recording, that no implausible values were present, and that interaction terms were correctly constructed as products of the relevant binary indicators.

Balance, Identification, and Robustness Checks

The Full-AI round reproduced all the original paper’s robustness checks in full including an F-Test on the equality of Post-Election and Post-Protest effects, re-estimation with expanded demographic controls, a subsample restricted to non-UR TV-watching respondents, and ordered logit specifications. All four original appendices were also reproduced, covering district-level respondent distributions, covariate balance across all pairwise period comparisons, partisan composition stability, non-response rates by institution, and daily trust trends. The AI-assisted round shifted focus toward methodological scrutiny, directing the AI to conduct code integrity checks and unit testing to verify that variables, period indicators, and interaction terms were correctly constructed. An omnibus balance test was run with standardized mean differences to assess whether respondents were systematically different across periods on observable characteristics. A McCrary density test at both event cutoffs probed whether the as-if random assignment of interview dates holds around each treatment threshold. Bonferroni and Benjamini-Hochberg corrections were applied across all institutional models to assess whether findings survive adjustment for multiple simultaneous hypothesis tests, and power analysis with minimum detectable effect calculations were run to evaluate whether the study, particularly the partisan subgroup models, was adequately powered to detect the effects it claims. A period-labeling bug was identified that reversed the direction of the descriptive comparisons which was caught in the comparison report on the first round, but after clearing the memory to run the AI-assisted round, persisted. A Corrected AI-assisted round addressed this, re-ran all analyses, and added three further checks. First was a day-of-week (DOW) adjustment to the McCrary density test, since interviewees likely worked more on weekdays than weekends, which could generate a spurious discontinuity at the cutoff independent of political events. We also implemented a December 5-6 sensitivity check restricting the post-election period to the two days before the protest was publicly announced on December 7, to test whether the paper's reported election effect is in fact party driven by anticipation effects. Lastly, a Period 1 and Period 3-only replication dropping all Period 2 respondents was run, isolating the cleanest possible pre-election versus post-protest contrast uncontaminated by the post-election window.

Full-AI Round: Replication Outcomes and Comparisons

Overall, the "Full AI" approach largely replicated the original findings, with one key but notable distinction. Through a number of tables and tests, the AI misattributed the periods (pre-election, post-election, post-protest) for portions of the data. Crucially, it did not merge data from different periods together (e.g. pre-election and post-election data being interpreted as belonging to one period and being analyzed together), resulting in replication results that were nearly entirely correct in magnitude and significance — some exceptions discussed below. However, this "swapping" of periods produced results that reversed the direction of the effect, suggesting in the final replication report that there was a post-protest trust decline.

For example, for Tables 11 and 12 in the Full AI Report, which paralleled the original Appendix II balance tests, the replication swapped the pre-election and post-election data, alongside the pre-election and post-protest data. Thus, the p-values were still correctly replicated, despite this swapping of periods. This error was present in Tables 1 (sample size), 4, and 5 (mean differences in political trust across periods) in the Full AI Report. The other notable discrepancy involves some results for the government trust index — specifically on Protest Day — which diverged more than the replication of other indices and periods. This resulted in the AI replication finding additional significant associations for this specific outcome (Table 6, Table 9.2, Table 16 in Full-AI Report).

Interestingly, when the LLM was prompted to compare its report with the original paper, it correctly identified the "swapping" discrepancy that resulted in a reversed main effect. In a subsection titled "A Critical Note on the Original Replication Report" (Full AI Comparison Report), it stated, "The narrative summary in our initial replication report incorrectly described the direction of the results, stating that trust 'declined' following the election and protest."

Replication Outcomes and Comparisons for AI-Guided Rounds

The AI-guided replication expanded on the previous replication by systematically evaluating Frye and Borisova's (2019) identifying assumptions, methodological choices, and statistical robustness. Unlike the full-AI round in which Claude determined its own analytical approach, these rounds were directed by specific prompts targeting a defined set of diagnostics (see Appendix X). The final Corrected AI-guided round additionally fixed the period-labeling bug discovered in the first round and subsequent Guided-AI stages, introduced a DOW control into a McCrary density test, restricted the post-election analysis to the two days preceding the protest announcement, and re-estimated the main models dropping Period 2 entirely.

The main OLS replication successfully reproduced the original regression coefficients for Tables 2 and 3 across both rounds (AI-Guided Report), confirming the findings are reproducible from the raw data. The governance index period3 coefficient of 0.158 matches the paper's reported 0.16, and the pattern of significant and non-significant outcomes in Table 2 is reproduced. In Table 3, the period3 main effects, United Russia (UR) baseline effects, and interaction term estimates all closely match the original paper's published values in magnitude and direction. After the period-labeling error, which mistakenly made trust appear to decline after the protest, is corrected in the third round, the descriptive Table 1 means also align with original findings. The governance index shows a broad and statistically significant trust increase of 0.242 ($p < 0.001$) from Period 1 to Period 3, consistent with the original paper's Table 1.

Notably, while the OLS estimates in Table 3 were successfully reproduced, the paper's statistical conclusion was not sustained once we applied multiple testing corrections. The original paper reports Table 3 results without any correction for multiple comparisons, which renders it susceptible to evidentiary threshold that may be overly lenient given the ten outcomes being tested simultaneously. Upon the addition of Bonferroni and Benjamini-Hochberg corrections, none of the interaction terms survive, suggesting the paper's interpretation of Table 3 as confirmatory evidence for the partisan updating mechanism is not fully supported.

Table 2: OLS regression with HC1 robust SEs. Dependent variable: institutional trust (1–5 scale, reversed). Reference category: Period 1 (pre-election). *** p

<table>
<tr><th>Outcome</th><th>period3</th><th>period2min</th><th>protestday</th><th>electday</th></tr>
<tr><td>trustgovind</td><td>0.158** (0.058)</td><td>|0.062 (0.067)</td><td>|0.177* (0.083)</td><td>|-0.043 (0.117</td></tr>
<tr><td>trustarmy</td><td>0.227** (0.079)</td><td>|0.282** (0.092</td><td>|0.504*** (0.11</td><td>) |-0.085 (0.164</td></tr>
<tr><td>trustpolice</td><td>0.285*** (0.069)</td><td>|0.242** (0.082</td><td>|0.377*** (0.10</td><td>) |0.222 (0.156)</td></tr>
<tr><td>trustfsb</td><td>0.273*** (0.076)</td><td>|0.185* (0.090)</td><td>|0.282* (0.123)</td><td>|0.211 (0.174)</td></tr>
<tr><td>trustcourt</td><td>0.130. (0.073)</td><td>|0.084 (0.086)</td><td>|0.201. (0.110)</td><td>|0.108 (0.154)</td></tr>
<tr><td>trustmungov</td><td>0.172* (0.073)</td><td>|-0.040 (0.084)</td><td>|0.138 (0.124)</td><td>|-0.207 (0.145</td></tr>
<tr><td>trustfedgov</td><td>0.174* (0.077)</td><td>|-0.054 (0.089)</td><td>|0.065 (0.122)</td><td>|-0.282. (0.15</td></tr>
<tr><td>trustduma</td><td>0.052 (0.078)</td><td>|-0.060 (0.087)</td><td>|-0.089 (0.123)</td><td>|-0.208 (0.141</td></tr>
<tr><td>trustprocuracy</td><td>-0.013 (0.076)</td><td>|-0.141 (0.088)</td><td>|-0.121 (0.121)</td><td>|-0.044 (0.149</td></tr>
<tr><td>trustun</td><td>0.051 (0.085)</td><td>|-0.106 (0.097)</td><td>|0.048 (0.126)</td><td>|-0.486** (0.1</td></tr>
</table>

Table 3: Partisan heterogeneity models (OLS, HC1 robust SEs). *** p

<table>
<tr><th>Outcome</th><th>period3</th><th>period2min</th><th>protestday</th><th>electday</th><th>ur</th><th>urafterprotest</th><th>urpostelection</th><th>apolitical</th><th>apolafterprotest</th><th>apolpostelection</th></tr>
<tr><td>trustgovind</td><td>0.195*
(0.080)</td><td>|0.052
(0.099)</td><td>|0.183*
(0.082)</td><td>|-0.002
(0.113</td><td>|0.559*** (0.</td><td>3. |-0.153
(0.</td><td>34. |-0.017
(0</td><td>162. |0.127
(</td><td>.080) |-0.011 (0</td><td>95. |0.055 (0</td></tr>
<tr><td>trustarmy</td><td>0.258*
(0.106)</td><td>|0.268*
(0.129</td><td>|0.496***
(0.11</td><td>) |-0.053
(0.159</td><td>|0.383* (0.15</td><td>) |-0.022 (0.</td><td>87. |0.078
(0.</td><td>38. |0.169
(</td><td>.111) |-0.051 (0</td><td>133. |0.023 (0</td></tr>
<tr><td>trustpolice</td><td>0.255**
(0.097)</td><td>|0.282*
(0.118</td><td>|0.384***
(0.10</td><td>) |0.263.
(0.149</td><td>|0.562*** (0.</td><td>40. |-0.089
(0.</td><td>77. |0.006
(0.</td><td>11. |0.139
(</td><td>.097) |0.112 (0.</td><td>20. |-0.
057
(</td></tr>
<tr><td>trustfsb</td><td>0.297**
(0.112)</td><td>|0.251.
(0.140</td><td>|0.288*
(0.122)</td><td>|0.254
(0.165)</td><td>|0.647*** (0.</td><td>43. |-0.187
(0.</td><td>89. |-0.188
(0</td><td>224. |0.143
(</td><td>.113) |0.036 (0.</td><td>36. |-0.
049
(</td></tr>
<tr><td>trustcourt</td><td>0.176.
(0.103)</td><td>|0.099
(0.132)</td><td>|0.192.
(0.111)</td><td>|0.143
(0.151)</td><td>|0.415** (0.1</td><td>3. |-0.184
(0.</td><td>82. |-0.086
(0</td><td>221. |0.170
(</td><td>.105) |-0.026 (0</td><td>126. |0.019 (0</td></tr>
<tr><td>trustmungov</td><td>0.241*
(0.101)</td><td>|-0.041
(0.123</td><td>|0.145
(0.120)</td><td>|-0.158
(0.151</td><td>|0.688*** (0.</td><td>32. |-0.187
(0.</td><td>71. |-0.145
(0</td><td>211. |0.088
(</td><td>.106) |-0.065 (0</td><td>127. |0.084 (0</td></tr>
<tr><td>trustfedgov</td><td>0.182.
(0.109)</td><td>|-0.113
(0.130</td><td>|0.087
(0.119)</td><td>|-0.234
(0.157</td><td>|0.671*** (0.</td><td>41. |-0.070
(0.</td><td>78. |0.157
(0.</td><td>9. |0.074
(</td><td>.109) |0.030 (0.</td><td>31. |0.1
26
(0</td></tr>
<tr><td>trustduma</td><td>0.146
(0.108)</td><td>|-0.112
(0.124</td><td>|-0.087
(0.119)</td><td>|-0.157
(0.146</td><td>|0.689*** (0.</td><td>58. |-0.260
(0.</td><td>93. |0.035
(0.</td><td>16. |0.122
(</td><td>.107) |-0.095 (0</td><td>129. |0.143 (0</td></tr>
<tr><td>trustprocuracy</td><td>0.059
(0.106)</td><td>|-0.125
(0.132</td><td>|-0.123
(0.123)</td><td>|0.001
(0.147)</td><td>|0.517*** (0.</td><td>50. |-0.251
(0.</td><td>90. |-0.157
(0</td><td>242. |0.186.</td><td>0.108) |-0.055 (0</td><td>129. |0.037 (0</td></tr>
<tr><td>trustun</td><td>0.043
(0.119)</td><td>|-0.033
(0.136</td><td>|0.059
(0.128)</td><td>|-0.476**
(0.1</td><td>8. |0.217
(0.162</td><td>|-0.151 (0.</td><td>15. |-0.253
(0</td><td>257. |-0.044</td><td>0.120) |0.077 (0.</td><td>50. |-0.
068
(</td></tr>
</table>

Key Diagnostic Findings

Balance and Exogeneity

Balance checks found significant imbalances between Period 1 and Period 3 on education (SMD = 0.260, $p < 0.001$) and the nopart indicator for apolitical respondents (SMD = 0.207, p = 0.003), with Period 3 containing a more educated and politically apathetic sample than the pre-election baseline (Section 10 in Corrected AI-Guided Report). For the Period 1 and Period 2 comparison, statesector (SMD = 0.157, p = 0.050) and nopart (SMD = 0.181, p = 0.031) were significantly imbalanced. Education and wealth imbalances were noted and partially controlled for within the original models, but the partisanship composition difference and the within-period timing problem are not addressed by the original paper. The within-period exogeneity test shows that covariates jointly predict interview timing in Period 1 ($R^2 = 0.08$, $F = 6.47$, $p = 0.0001$), driven in part by a strong effect of wealth, while Periods 2 and 3 pass the test with low R^2 and insignificant p-values.. Further, the prompted MANOVA balance test failed to run in the AI-guided round because two variables were perfectly redundant with others, making the covariance matrix singular and preventing estimation. After dropping these redundant variables, the test runs properly and shows that the groups are not balanced. In substantive terms, all period comparisons are statistically different, meaning the samples are not comparable on observables.

McCrary Density Tests

A standard McCrary density test at the election cutoff produced a robust test statistic of -2.43 ($p = 0.015$), suggesting a significant discontinuity (Section 11 in Corrected AI-Guided Report). The DOW adjusted specification introduced in round three, which residualizes daily interview counts on weekday dummies before testing for a discontinuity, reduced this to a non-significant result with a discontinuity coefficient of -12.881 ($SE = 13.956$, $p = 0.380$), indicating that the initial discontinuity was likely driven by weekly fieldwork patterns. At the protest cutoff, both the unadjusted discontinuity of 1.498 ($p = 0.959$) and the DOW-adjusted discontinuity of 10.316 ($p = 0.577$) were non-significant. These results are consistent with quasi-random assignment assumption at both cutoffs once fieldwork schedules were taken into account. However, the DOW model explains only 13.1 percent of variance in daily counts, and the large standard errors at the protest cutoff mean the test has limited power to detect a discontinuity of plausible magnitude.

Power Analysis

The additional power analysis found that the main Period 1 versus Period 3 comparison on the governance index achieved 82.6 percent power at alpha = 0.05 and fell to 53.7 percent at the Bonferonni-corrected threshold of alpha = 0.005 (Section 12 in Corrected AI-Guided Report).

Several individual outcomes including trust in courts (46.9 percent), the Duma (11.4 percent), procuracy (5.4 percent), and the United Nations (10.3 percent) were underpowered even at the uncorrected threshold. The minimum detectable effect for the urafterprotest interaction term was 0.382 scale units, which is nearly double the magnitude of any observed coefficient. This means the study was insufficiently powered to test the partisan mechanism proposed in Table 3. The Period 1 versus Period 2 comparison was insufficiently powered to test the partisan mechanism proposed in Table 3. This comparison was the weakest in the study, given its smaller comparison group, and achieved only 69.7 percent power to detect an effect the size of the main Period 3 governance-index coefficient, and it would be even more underpowered for the smaller observed post-election effect.

Anticipation Effects and December 5 to 6 Robustness Check

Our test for anticipation effects (Section 13 in Corrected AI-Guided Report) found that statistically significant post-election effects on the governance index were driven primarily by December 5 and 6 (beta = 0.217, p = 0.022), disappearing from December 7-9 (beta = 0.072, p = 0.413). For federal and municipal government trust, the post-election coefficient reversed sign between the pre- and post-protest announcement windows. The December 5 to 6 robustness check showed that replacing the post-election indicator with one covering only December 5 and 6 produces a governance index post-election coefficient of 0.245 (p = 0.002), compared to 0.062 (non-significant) in the original specification. The pattern holds across eight of ten outcomes, with trust in the Duma increasing from -0.060 to 0.259 (p = 0.009), federal government from -0.054 to 0.282 (p = 0.006), and municipal government from -0.040 to 0.272 (p = 0.005). Importantly, this does not support our original hypothesis that protest anticipation inflates the estimated election effect. The December 7-9 data attenuates, and in some cases, reverses the positive post-election signal, suggesting the baseline specification is underestimating the election's impact. This raises a fundamental concern about temporal instability within Period 2 since the post-election treatment effect is highly sensitive to the choice of window, and the December 5-9 period cannot be treated as a homogenous post-election interval.

Documentation of Discrepancies

The most significant discrepancy across all rounds was the period-labeling error. The helper function assigned the reference-period label to observations when the period indicator equaled zero, while the original paper assigned period indicators equal to one. Groups were then swapped in all pairwise comparisons, producing differences correct in magnitude but inverted in sign. Unit tests in the Corrected AI-guided round formally documented and confirmed the bug, and the corrected function resolved the issue. The OLS regression models were unaffected, as those models use period indicators as regressors instead of grouping variables.

Other discrepancies identified through the extended diagnostics include the balance failures on education and apolitical respondent composition between Periods 1 and 3, the Period 1 exogeneity failure driven by wealth, and the internal heterogeneity of the post-election window, which the paper did not explicitly acknowledge. Education and wealth imbalances are controlled in the original models, but the partisanship composition difference and the within-period timing problems are unaddressed. The period definition instability in the December 5 to 6 robustness check is the most consequential discrepancy. The paper's conclusion that the election had little systematic effect on trust is a product of including post-announcement days in the post-election window, and the paper's claim to have cleanly separated election and protest effects rests on a period definition whose internal heterogeneity it does not test.

Further, in the original specification, the omnibus MANOVA balance test failed to produce results due to a rank-deficiency caused by perfectly collinear variables. After repairing this issue in the Corrected AI-guided round, the MANOVA indicates statistically significant differences in covariates across all survey periods, which was not reported in the original analysis. Thus, the identifying assumption of covariate balance is weaker than implied in the original study. Covariate imbalance implies that changes in trust may partly reflect differences in respondents across periods, not the causal impact of the election or protest. However, the small Pillai trace values (0.023 to

0.055) indicate that the effect size of the imbalances are modest.

Critical Analysis

The AI was highly effective at structured diagnostic tasks once directed by specific prompts. It correctly identified the period-labeling bug through unit testing, produced accurate balance checks, ran appropriate exogeneity tests, and implemented multiple testing corrections appropriately. The anticipation effects analysis and December 5 to 6 robustness checks were well executed and produced interpretable results. The AI was also effective at translation tasks, accurately converting Stata code to R to reproduce the original models with correct standard error specifications, and implemented additional checks cleanly once directed. It also handled the conceptual framing of diagnostics appropriately, correctly identifying the anticipation contamination concern and designing a split test around the December 7 announcement.

However, the AI's performance on autonomous error detection was subpar. The period-labeling error was present in both the Full-AI round and the AI-guided round, persisting across two independent runs before being manually flagged in the Corrected AI-guided round. The AI consistently reproduced its own mistake even after clearing the memory, suggesting a crucial limitation for AI replication. It inconsistently flagged this discrepancy which directly contradicted the paper's findings, and only reliably addressed it when explicitly prompted to run unit tests. The AI also handled the aforementioned omnibus MANOVA test poorly, which the AI did not flag before rendering the output. This resulted in a table of NA values appearing in the AI-guided report without explanation. This suggests AI replication tools are reactive, not proactive in error detection, requiring researcher-directed verification to catch these kinds of systematic mistakes.

Several changes would significantly improve the reproducibility of this paper. A pre-specification of the correction family for multiple testing and reporting corrected p-values alongside raw ones would make the evidentiary standards more transparent. A power analysis conducted prior to data collection, particularly for the Table 3 subgroup heterogeneity analysis, would clarify what the study could realistically detect. If standard errors were date-clustered rather than

HC1 robust, this would address within-day correlation in errors. Lastly, testing the Period 2 window definition would strengthen the paper's claim to have identified distinct election and protest effects.

Conclusion

In sum, the AI replication confirms that Frye and Borisova's (2019) core findings are replicable. The post-protest trust increase survives across all three rounds, multiple testing corrections, and the removal of Period 2 from the estimation sample, and the OLS coefficients are successfully reproduced. However, the extended diagnostics reveal a more complicated narrative. The partisan heterogeneity and updating mechanism in Table 3 lacks confirmatory statistical power after multiple testing, and the as-if random assumption is only partially satisfied for period 1, where wealth significantly predicts interview timing within the pre-election period. The most consequential finding for our replication is that restricting Period 2 to December 5 and 6 reveals that the contaminated post-announcement days included in the pre-protest period attenuate the post-election signal, suggesting that period ranges shape the paper's conclusion that elections had little systematic effect on trust. Finally, the corrected MANOVA balance tests indicate statistically significant, but modest, differences across periods, suggesting that while imbalance is present, it is limited and may only partially contribute to the estimated effects.

The most important methodological lesson from this paper concerns the limits of AI autonomy in quantitative research. AI-assisted replication is most effective when it is treated as a collaborative project between researchers and AI models. Following from this project, researchers should structure their prompts around specific and theoretically motivated diagnostics rather than open-ended tasks. Many of the most important findings in this replication required human reasoning to interpret correctly. Future interactions of this replication should also consider pursuing additional diagnostics to strengthen the McCrary density test at the protest cutoff, specifically a bandwidth sensitivity analysis that tests whether the non-significant discontinuity holds across progressively wider windows, a power calculation establishing the minimum detectable discontinuity given available daily interview counts, and a placebo cutoff test comparing the protest cutoff result against arbitrary dates where no event occurred. These tests may help distinguish between an absence of sorting and a null result driven by insufficient statistical power in the local window. Taken together, the three rounds of replication demonstrate that AI tools can meaningfully extend the scope of quantitative replication beyond simple coefficient reproduction, but only when researchers are actively involved in specifying what to test, verifying what the AI produces, and interpreting what the results mean for the paper's theoretical claims.

References

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